SRDE / Blackbird: prior research and experiment novelty audit

3 November 2026. Research direction: Dan Merluse. Source retrieval, comparison and documentation are AI-assisted.

Decision

The completed studies contain substantial established material. Their actual results remain useful, but no unique recursive-portal method or new logic has been established. They should be described as SRDE-specific diagnostics, benchmarks and controlled comparisons. Different seeds, renamed graph traversal, a different reset threshold or an equivalent recursive implementation do not qualify as new scientific contributions.

The proposed B09 false-alarm calibration run is rejected as a standalone original study. It can serve as engineering or a conventional control within a genuinely distinct question. No new simulation was launched during this literature audit.

What was checked

This targeted review records 18 primary studies spanning 1977–2026. Searches used web search, Firecrawl and Consensus discovery, then publisher, author and conference sources. Read scopes are recorded per source; several records are abstract-level. This is not a complete systematic review, patent clearance, proof of global uniqueness, or independent peer review. The audit compares the eight recovered completed study collections; it does not adjudicate another chat's active P3 study. E28 remains unavailable.

Discovery groups: four-valued evidence and incomplete/conflicting information; symbol grounding and valence; provenance and truth maintenance; path-based knowledge-graph inference; drift and recurring conditions; delayed labels and evaluation; active feature/source acquisition; uncertainty under shift. Source dates are publication dates rather than misleading web crawl dates. S13 is first online in 2025 but its recovered proceedings chapter bears 2026 copyright; S08's inspected extended draft is 2006 for the 2007 publication; S16 is a 2012 volume presented at VLDB 2013.

Closest overlaps and consequences

S01: A Useful Four-Valued Logic

Belnap (1977). Distinguishing inconsistent and incomplete evidence has a long formal history. SRDE consequence: Binary evidence states are an implementation/diagnostic, not a new logic. Read scope: publisher abstract. Primary source.

S02: The Symbol Grounding Problem

Harnad (1990; arXiv deposit 1999). Arbitrary symbols require grounding beyond further arbitrary symbols. SRDE consequence: Glyph relabeling and digit classification do not solve general grounding. Read scope: author abstract; journal metadata. Primary source.

S03: A Circumplex Model of Affect

Russell (1980). Pleasantness and arousal are established affect dimensions. SRDE consequence: F in [0,1] is a coding convention, not novel affect theory or truth. Read scope: primary PDF, introductory material. Primary source.

S04: Provenance Semirings

Green, Karvounarakis, Tannen (2007). Annotations track derivations; recursive provenance has algebraic and fixed-point treatments. SRDE consequence: Provenance plus recursion alone is not a novel contribution. Read scope: primary PDF, abstract and recursive Datalog sections. Primary source.

S05: An assumption-based TMS

de Kleer (1986). Assumption sets support reasoning across inconsistent alternatives. SRDE consequence: Context/dependency tracking requires comparison with truth-maintenance approaches. Read scope: primary PDF and publisher abstract. Primary source.

S06: Go for a Walk and Arrive at the Answer (MINERVA)

Das et al. (ICLR 2018; preprint 2017, v2 2018). Learned query-conditioned graph traversal predates SRDE; evaluation includes leakage precautions. SRDE consequence: Path traversal, multi-hop inference and unseen relations are established. Read scope: primary full-text methods and evaluation passages. Primary source.

S07: Neural Bellman-Ford Networks

Zhu et al. (2021). Generalized path products and sums are learned through Bellman-Ford-style computation. SRDE consequence: Recursive path composition needs an operational difference from established graph inference. Read scope: primary PDF introduction and formulation. Primary source.

S08: Learning from Time-Changing Data with Adaptive Windowing

Bifet, Gavalda (2007 publication; author draft 17 October 2006). Adaptive windows address drift with statistical error bounds. SRDE consequence: A better reset threshold, decay or window is not itself a fresh research problem. Read scope: primary extended draft, introduction and algorithm passages. Primary source.

S09: Bayesian Online Changepoint Detection

Adams, MacKay (2007). Online run-length posterior and predictive distributions handle abrupt changes. SRDE consequence: Bayesian state/change uncertainty must be included among conventional alternatives. Read scope: primary full-text algorithm and evaluation. Primary source.

S10: Change with Delayed Labeling: When is it Detectable?

Zliobaite (2010 conference; indexed 2011). Studies which changes can be detected before labels arrive. SRDE consequence: Label delay and detectability are established topics; detailed method comparison remains open. Read scope: publisher abstract only. Primary source.

S11: Delayed labelling evaluation for data streams

Grzenda, Gomes, Bifet (online 2019; volume 2020). Evaluation must respect label latency; prediction refinement can be evaluated over time. SRDE consequence: Timestamp guards are necessary evaluation practice, not evidence of originality. Read scope: primary full-text abstract and introduction. Primary source.

S12: Active feature acquisition on data streams under feature drift

Beyer et al. (2020). Selects whether and which missing feature to acquire under cost and changing streams. SRDE consequence: Budgeted adaptive evidence collection under drift already exists. Read scope: primary full-text method section. Primary source.

S13: Investigating Delays of Combined Feature and Label Acquisitions on Data Streams

Buttner, Spiliopoulou (first online 5 October 2025; proceedings copyright 2026). Combines feature and label queries, costs, missingness and delayed responses on streams. SRDE consequence: Adding delayed queries to acquisition is insufficient novelty. Full chapter recovered from conference proceedings. Read scope: publisher abstract and conference-hosted full chapter, methods/evaluation/conclusion. Primary source.

S14: Conformal Inference for Online Prediction with Arbitrary Distribution Shifts

Gibbs, Candes (2024). Adapts prediction sets to distribution shift. SRDE consequence: Online uncertainty adjustment is established; probability calibration and set coverage are distinct. Read scope: publisher abstract and primary PDF retrieved. Primary source.

S15: Truth Discovery and Copying Detection in a Dynamic World

Dong, Berti-Equille, Srivastava (2009). Models source freshness, quality, evolving copying and changing true values. SRDE consequence: Temporal provenance-aware truth discovery is especially close prior art. Read scope: primary PDF, model and evaluation passages. Primary source.

S16: Less is More: Selecting Sources Wisely for Integration

Dong, Saha, Srivastava (2012 publication; VLDB 2013 presentation). Trades information integration quality against source acquisition and integration costs. SRDE consequence: Selecting fewer useful sources and considering marginal value is established. Read scope: primary PDF introduction and formulation. Primary source.

S17: Active multi-fidelity Bayesian online changepoint detection

Gundersen et al. (2021). Selects measurement fidelity through information gain per cost for change detection. SRDE consequence: Cost-sensitive sensing plus drift is not a fresh combination by itself. Read scope: primary PDF method formulation. Primary source.

S18: Learning-To-Measure: In-Context Active Feature Acquisition

Kobayashi et al. (2026). Acquires features using uncertainty and conditional mutual information under missingness and scarce labels. SRDE consequence: Recent acquisition methods belong in the comparison search; detailed implementation review remains open. Read scope: proceedings abstract only. Primary source.

Experiment-by-experiment decisions

Portal Hypothesis v0.1: established ingredients; SRDE diagnostic. More levels or a portal name is not a substantive novelty claim. Closest source IDs: S04 S06 S07.

P1 blind worlds: controlled comparison / equivalence diagnostic. New generated seeds are replications; equal computations cannot establish an architecture advantage. Closest source IDs: S04 S06 S07.

P2 historical CO2: application benchmark; no method novelty established. A new dataset can test external validity but does not establish a new forecasting algorithm. Closest source IDs: S08 S09 S11.

Unseen-data pilot: pilot on an established task. Use strong graph baselines and an official hard-negative/evaluation protocol before a new empirical claim. Closest source IDs: S06 S07.

Binary semantics: diagnostic ablations of established distinctions. Keep bit storage, truth, activation, uncertainty and pleasantness distinct; richer coding is not novel logic. Closest source IDs: S01 S02 S03 S04 S05 S15.

Learned temporal: SRDE-specific comparison; broad premise established. Separate source error, state dynamics, copying and freshness rather than merely changing drift strength. Closest source IDs: S08 S09 S12 S15.

Delayed feedback: bounded comparison; no novelty claim cleared. Whole-stream losses are useful evidence, but delayed adaptation and recurring conditions are prior research areas. Closest source IDs: S08 S09 S10 S11 S12 S13 S14 S15 S17.

Historical E30: implementation benchmark; literature gap not cleared. A comprehensive retrieval/memory benchmark review is still needed before claiming originality for E30. Closest source IDs: S04 S05 S06.

Proposed B09 detector calibration: REJECTED as a standalone novel contribution. Useful engineering baseline/control only; this is not approved as the next unique SRDE study. Closest source IDs: S08 S09 S10 S11 S17.

Proposed provenance audit intervention: CANDIDATE EXTENSION; NOT NOVELTY-CLEARED. Must close targeted audit/lineage acquisition literature and distinguish from multi-fidelity sensing and dynamic copying inference. Closest source IDs: S04 S05 S12 S13 S15 S16 S17 S18.

Candidate direction, not a cleared originality claim

The candidate asks when a costly, delayed, fallible audit of source lineage is more useful than another assertion, while reports before that intervention are deliberately matched. It separates a change in copying relationships from a change in the target and charges every method for the information it obtains. Our earlier simulator supplied lineage identity; this candidate would make uncertainty about that identity a first-class decision problem.

This is an inference about a potentially useful extension, not a discovery that nobody has studied it. Dynamic copying inference (S15), source selection (S16), multi-fidelity sensing (S17), and combined delayed acquisition (S13) are already close. The candidate remains NOT CLEARED until targeted active truth-discovery/causal-sensing literature and recent acquisition methods are compared in detail. It is not preregistered and has no experimental results. The precise primary estimand and cost scale must be frozen before execution. No claim that merely combining these ingredients is novel is accepted.

Procedure adopted for future work

NOVELTY_GATE.md is the required project checklist. Before each discovery run, the record must name the nearest studies, the internal predecessor, the exact material difference and a falsifiable reason it matters. Closest published methods become baselines where applicable. Literature-derived guarantees do not transfer automatically to a changed SRDE setting. Timestamp guards and reproducibility are quality requirements, not novelty.

Necessary replications and negative controls are explicitly labeled. Originality is required of the new research question, not of every control arm. If a close match is found, the next run becomes a labeled reproduction or is redesigned before its protocol is frozen. Existing evidence is never erased to manufacture originality.

Retrieval limits and recovery

The first Green PDF retrieval timed out and its institutional repository returned 403; the author PDF was subsequently retrieved. An initial VLDB static-dependence PDF route failed; the dynamic-world author PDF was retrieved and is the stronger temporal comparison used here. S10 remains publisher-abstract only. S18 remains proceedings-abstract only. The S13 publisher chapter preview was insufficient for a method-level contrast; the conference's publicly hosted proceedings were downloaded locally and the relevant chapter methods/evaluation/conclusion were inspected. The full third-party book and PDFs are not republished in this package.

Consensus supplied a recurring-drift survey discovery record, but its fetch lacked an abstract, so it is not counted as a read primary study or used for a detailed claim. No citation-count ranking is used as evidence of novelty. The raw search trail is summarized in SEARCH_LOG.json; no exhaustive database coverage or automatic host-wide enforcement is claimed.